Task 1 & 2 Write-Up: Adversity Profile and Client Profitability

Task 1 — Adversity Profile

A trade is labelled *adverse* at horizon τ if the LP's PnL at that horizon is negative: **side × (M_τ – TP) < 0**. Adversity percentage is the fraction of a client's trades that are adverse.

Client	τ=5	τ=10	τ=15	τ=20	τ=25	τ=30
A	39.7%	41.4%	42.2%	41.9%	41.9%	41.7%
B	40.8%	42.4%	43.2%	43.3%	43.4%	43.3%
C	41.4%	43.3%	44.2%	44.8%	45.4%	45.9%
D	43.3%	46.5%	48.2%	49.7%	50.6%	51.8%
E	45.0%	48.5%	50.9%	52.1%	53.8%	54.8%
F	47.0%	51.7%	54.3%	56.9%	59.5%	61.9%

Table 1: Adversity percentage (%) per client per horizon τ (seconds).

Trends and Interpretation

- **Clients A and B — Noise traders (flat profile):** Adversity is low (~40–43%) and nearly constant across horizons. Price moves weakly against the LP over 30 s, suggesting these clients trade for liquidity reasons, not information. Their flow is noise-driven.
- **Client C — Mild slope:** Adversity climbs slowly from 41% to 46%. There is a slight informational edge — trades are marginally predictive of short-term price direction — but the LP still earns a positive spread margin.
- **Client D — Moderate slope:** Rises from 43% at τ=5 to 52% at τ=30. Crosses the 50% breakeven mark at τ≈25. The client's trades contain more directional information, particularly at longer horizons, indicating semi-informed order flow.
- **Client E — Adverse slope, costly:** Starts at 45% but reaches 55% by τ=30. The LP loses money in aggregate, meaning this client's trades consistently move in the direction the client anticipated.
- **Client F — Highly informed / toxic flow:** The steepest profile, reaching 62% at τ=30. This is consistent with a client trading on genuine short-lived private information. The LP is systematically on the wrong side by a large margin at longer horizons.

The **monotone increase with** τ across all clients (especially D, E, F) is consistent with the adverse-selection literature: informed traders' informational advantage is realised over time, while noise traders' trades are uncorrelated with future price moves.

Task 2 — Client Profitability and Spread Recommendation

Expected PnL per trade at horizon τ is **E[side × V × (M_τ – TP)]** (eq. 5). Aggregate PnL uses uniform weights across all six horizons (eq. 6).

Client	τ=5	τ=10	τ=15	τ=20	τ=25	τ=30	Agg PnL	Class.	δ*
A	1.27	1.54	1.74	2.06	2.29	2.49	+1.90	Profitable	0.0000
B	1.21	1.34	1.54	1.73	1.91	2.06	+1.63	Profitable	0.0000
C	1.08	1.16	1.19	1.16	1.23	1.22	+1.17	Profitable	0.0000
D	0.87	0.68	0.49	0.18	–0.08	–0.35	+0.30	Profitable	0.0096
E	0.65	0.30	–0.11	–0.57	–1.07	–1.61	–0.40	Costly	0.0190
F	0.45	–0.18	–0.92	–1.75	–2.61	–3.63	–1.44	Costly	0.0329

Table 2: Expected PnL per trade at each τ, aggregate PnL, classification, and minimum half-spread δ^* (all in price units).

Classification Justification

Classification is based on the sign of aggregate PnL (eq. 6), which weights all six closing horizons equally. **A, B, C, D are net-profitable** to trade with — their aggregate PnL is positive, meaning the LP earns more from the spread than it loses to adverse moves on average. **E and F are costly**: their aggressive adversity profiles (55% and 62% at τ=30) generate systematic losses that overwhelm any spread income at current quoted levels.

Minimum Half-Spread δ^* and its Relation to Adversity

When the LP quotes at $M_0 \pm \delta$, the effective TP shifts to $TP_{new} = M_0 - side \times \delta$. The aggregate PnL becomes:

$$E[side \times V \times avg_i(M_{ti} - M_0)] + \delta \times E[V]$$

Setting this ≥ 0 gives δ*** = max(0, –E[side·V·avg(M_{ti}–M_0)] / E[V])**. Clients with higher adversity profiles require larger δ^* because the adverse price drift (negative term in numerator) is larger. Specifically:

- A, B, C: δ^* = 0. Already profitable at M0 — any positive spread earns more.
- D: δ^* = 0.0096. Modest adverse drift at long horizons; small premium suffices.
- E: δ^* = 0.0190. Strong adverse drift from τ=15 onward demands ~2× D's premium.
- F: δ^* = 0.0329. Highly toxic flow; the required spread approaches the full market half-spread (~0.015), making F borderline uneconomic to trade.

The monotone ordering **δ^*(A)=δ^*(B)=δ^*(C) < δ^*(D) < δ^*(E) < δ^*(F)** mirrors precisely the ordering of adversity profiles, confirming that δ^* is a direct monetisation of the adversity slope.